LETTERS
https://doi.org/10.1038/s41567-018-0189-6

nature physics

Electrically switchable Berry curvature dipole in the monolayer topological insulator WTe_2

Su-Yang Xu[1,8], Qiong Ma[1,8], Huitao Shen[1], Valla Fatemi[1], Sanfeng Wu[1], Tay-Rong Chang[2], Guoqing Chang[3], Andrés M. Mier Valdivia[1], Ching-Kit Chan[4], Quinn D. Gibson[5], Jiadong Zhou[6], Zheng Liu[6], Kenji Watanabe[7], Takashi Taniguchi[7], Hsin Lin[3], Robert J. Cava[5], Liang Fu[1], Nuh Gedik[1]* and Pablo Jarillo-Herrero[1]*

Recent experimental evidence for the quantum spin Hall (QSH) state in monolayer WTe_2 has linked the fields of two-dimensional materials and topological physics[1–7]. This two-dimensional topological crystal also displays unconventional spin–torque[8] and gate-tunable superconductivity[7]. Whereas the realization of the QSH has demonstrated the nontrivial topology of the electron wavefunctions of monolayer WTe_2, the geometrical properties of the wavefunction, such as the Berry curvature[9], remain unstudied. Here we utilize mid-infrared optoelectronic microscopy to investigate the Berry curvature in monolayer WTe_2. By optically exciting electrons across the inverted QSH gap, we observe an in-plane circular photogalvanic current even under normal incidence. The application of an out-of-plane displacement field allows further control of the direction and magnitude of the photocurrent. The observed photocurrent reveals a Berry curvature dipole that arises from the nontrivial wavefunctions near the inverted gap edge. The Berry curvature dipole and strong electric field effect are enabled by the inverted band structure and tilted crystal lattice of monolayer WTe_2. Such an electrically switchable Berry curvature dipole may facilitate the observation of a wide range of quantum geometrical phenomena such as the quantum nonlinear Hall[10,11], orbital-Edelstein[12] and chiral polaritonic effects[13,14].

One of the early landmarks of condensed matter physics was the classification of metals, insulators and semiconductors by studying the energy–momentum dispersion—or band structure—of the electrons in crystalline solids. Despite this remarkable success, the quantum nature of the electron states means that they can only be fully described by their quantum wavefunctions, whereas the band structure concerns only the energy and momentum eigenvalues of the wavefunctions. Therefore, a central question in modern condensed matter physics is whether there exist new phenomena that arise from other properties of quantum wavefunctions beyond the band structure[9]. For instance, the study of the global—or topological—properties of electronic wavefunctions continues to give rise to novel topological phases, including the QSH states, three-dimensional topological insulators and Weyl semimetals. These topological materials feature robust surface states and often exhibit protected transport and optical responses. Another direction is to study the local—or geometrical—properties of wavefunctions. One important property is the local curvature of the wavefunction, defined as the Berry curvature. Originally employed to explain the anomalous Hall conductivities of ferromagnets[9], the importance of the Berry curvature is increasingly recognized in a wide range of areas in condensed matter physics, including nonlocal transport and chiral optical responses in noncentrosymmetric metals and semiconductors[10–12,15–20], unconventional pairing in superconductors[21], and topological plasmonic and excitonic polaritons[13,14]. Moreover, the interplay between topology and Berry curvature, although of great fundamental interest, has only rarely been explored. This is because most topological materials have zero Berry curvature in their bulk electronic states due to inversion symmetry, whereas materials with nonzero Berry curvature (for example, monolayer MoS_2 or gapped graphene) are usually topologically trivial. In general, research on understanding the effects of quantum geometry in novel materials is developing rapidly both in theory and in experiments[10–21]. It is important to find new materials with novel quantum geometrical and topological properties.

Bulk WTe_2 crystals were found to show a large, non-saturating magnetoresistance[22] and were proposed as type-II Weyl semimetals[23]. Spin–torque and gate-tunable superconductivity were also observed in monolayer WTe_2[7,8], suggesting nontrivial spin-resolved Berry curvature properties in WTe_2. Meanwhile, although monolayer WTe_2 has been studied by electronic transport[3,6,7], angle-resolved photoemission[4], and scanning tunnelling microscopy[4,5], optical studies are currently lacking.

The circular photogalvanic effect (CPGE), also noted as the injection current[24], is the generation of electrical currents whose direction switches on reversing the chirality of light. Previous CPGE experiments on semiconductor heterostructures[25], topological insulators[26,27] and semiconducting transition metal dichalcoginides[28] have attracted great interest because of the ability to optically generate spin-polarized electrical currents. On the other hand, a systematic microscopic mechanism of these CPGEs is usually difficult to achieve because the complex optical processes often involve many bands. However, recent theoretical advances[19] have shown that when the inter-band transition involves only the lowest two bands, the CPGE in three-dimensional bulk materials has a concise microscopic origin that arises from the nontrivial Berry curvature.

[1]Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA. [2]Department of Physics, National Cheng Kung University, Tainan, Taiwan. [3]Institute of Physics, Academia Sinica, Taipei, Taiwan. [4]Department of Physics and Astronomy, University of California Los Angeles, Los Angeles, CA, USA. [5]Department of Chemistry, Princeton University, Princeton, NJ, USA. [6]Centre for Programmable Materials, School of Materials Science and Engineering, Nanyang Technological University, Singapore, Singapore. [7]National Institute for Materials Science, Tsukuba, Japan. [8]These authors contributed equally: Su-Yang Xu, Qiong Ma. *e-mail: gedik@mit.edu; pjarillo@mit.edu

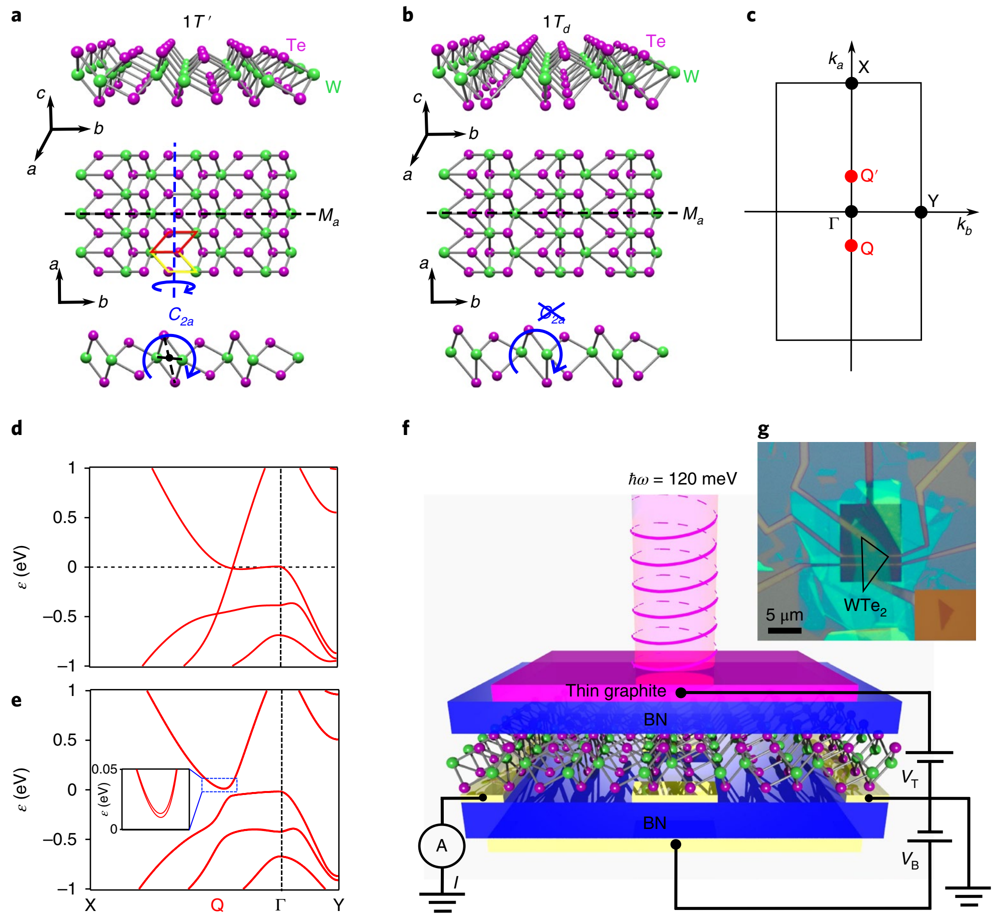

Fig. 1 | Crystal and electronic structures of monolayer WTe_2. **a**, The 1T' structure of monolayer WTe_2 consists of a mirror plane $\mathcal{M}_a$ and a screw rotation symmetry C_{2a}. C_{2a} is nonsymmorphic: it involves a 180° rotation about $\hat{\mathbf{a}}$ and a $\frac{a}{2}$ translation along $\hat{\mathbf{a}}$, as depicted by the yellow and red parallelograms. **b**, The 1T_d structure has only the mirror plane $\mathcal{M}_a$. The rotational symmetry C_{2a} is broken (exaggerated). **c**, The first Brillouin zone with important momenta labelled. **d**,**e**, Band structure of monolayer WTe_2 without and with spin-orbit coupling, respectively. **f**, Schematic experimental set-up for detecting the mid-infrared circular photogalvanic effect on a dual-gated monolayer WTe_2 device. **g**, Optical image of a dual-gated monolayer WTe_2 device. Scale bar: 5 µm.

As a result, aside from being an interesting and potentially useful phenomenon, the CPGE under such conditions becomes a powerful probe of important wavefunction properties of a bulk material, such as the chirality and the topological charge of the Weyl nodes[19,29]. Here, we show that the CPGE under the same conditions in two dimensions also has a clear origin in the Berry curvature.

The monolayer WTe_2 lattice can be described by two possible structural phases, 1T' and 1T_d. The inversion-symmetric 1T' structure has been widely assumed in previous work[1–4,6]. This structure has two independent symmetries: the mirror symmetry $\mathcal{M}_a$ and the two-fold screw rotational symmetry C_{2a}(Fig. 1a), whose combination gives rise to the inversion symmetry of the 1T' phase. The 1T_d structure (Fig. 1b), which is defined here as the monolayer directly isolated from the inversion-breaking bulk T_d WTe_2 lattice structure (see details in Supplementary Information VII.3), deviates slightly from 1T'. In 1T_d, $\mathcal{M}_a$ is preserved but C_{2a} is weakly broken. As a result, 1T_d actually breaks inversion symmetry, which affects its electronic structure. The low-energy band structure of monolayer WTe_2 without spin–orbit coupling features tilted two-dimensional Dirac fermions at the Q and Q′ points (Fig. 1c,d)[30]. The inclusion of spin–orbit coupling leads to an inverted, indirect QSH gap, where the valence band top is located at the Γ point and the conduction band bottom is located at the Q and Q′ points. The weak inversion symmetry breaking of 1T_d further induces a small spin splitting near the bottom of the conduction band (Fig. 1e inset).

Here we use a mid-infrared scanning photocurrent microscope equipped with a CO_2 laser ($\lambda = 10.6\,\mu$m, $\hbar\omega \simeq 120$ meV) to detect the CPGE induced by the inter-band transition across the inverted QSH gap near Q and Q′. We have fabricated high-quality, encapsulated, dual-gated monolayer WTe_2 devices with multiple electrodes (Fig. 1f,g). The dual gates allow us to independently vary the charge density n and the displacement field $\mathbf{D}$ (see details in the Methods section).

We first present our data at $T = 150$ K without external displacement fields. Figure 2a,c shows the measured photocurrents along two orthogonal directions ($I_{\hat{a}}$ and $I_{\hat{b}}$) as a function of the beam spot position. The photocurrent changes sign as one moves the beam spot from one contact to the opposite. This bipolar spatial pattern reveals the photo-thermal current[27,28] along both $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$, which is due to the different Seebeck coefficients of WTe_2 and the metal contacts (Supplementary Information I.3). In contrast, the polarization dependence of the photocurrent along the two directions (Fig. 2b,d) is distinctly different. $I_{\hat{a}}$ (Fig. 2d) shows a significant modulation with light polarization, which reaches its maximum for right-circularly polarized light, its minimum for left-circularly polarized light, and is zero for linearly polarized light. This pattern clearly demonstrates the existence of the CPGE along $\hat{\mathbf{a}}$. $I_{\hat{b}}$, on the other hand, shows no observable dependence on polarization (Fig. 2b). To further distinguish the CPGE from the photo-thermal effect, we study the dependence with charge density, without applying displacement fields.

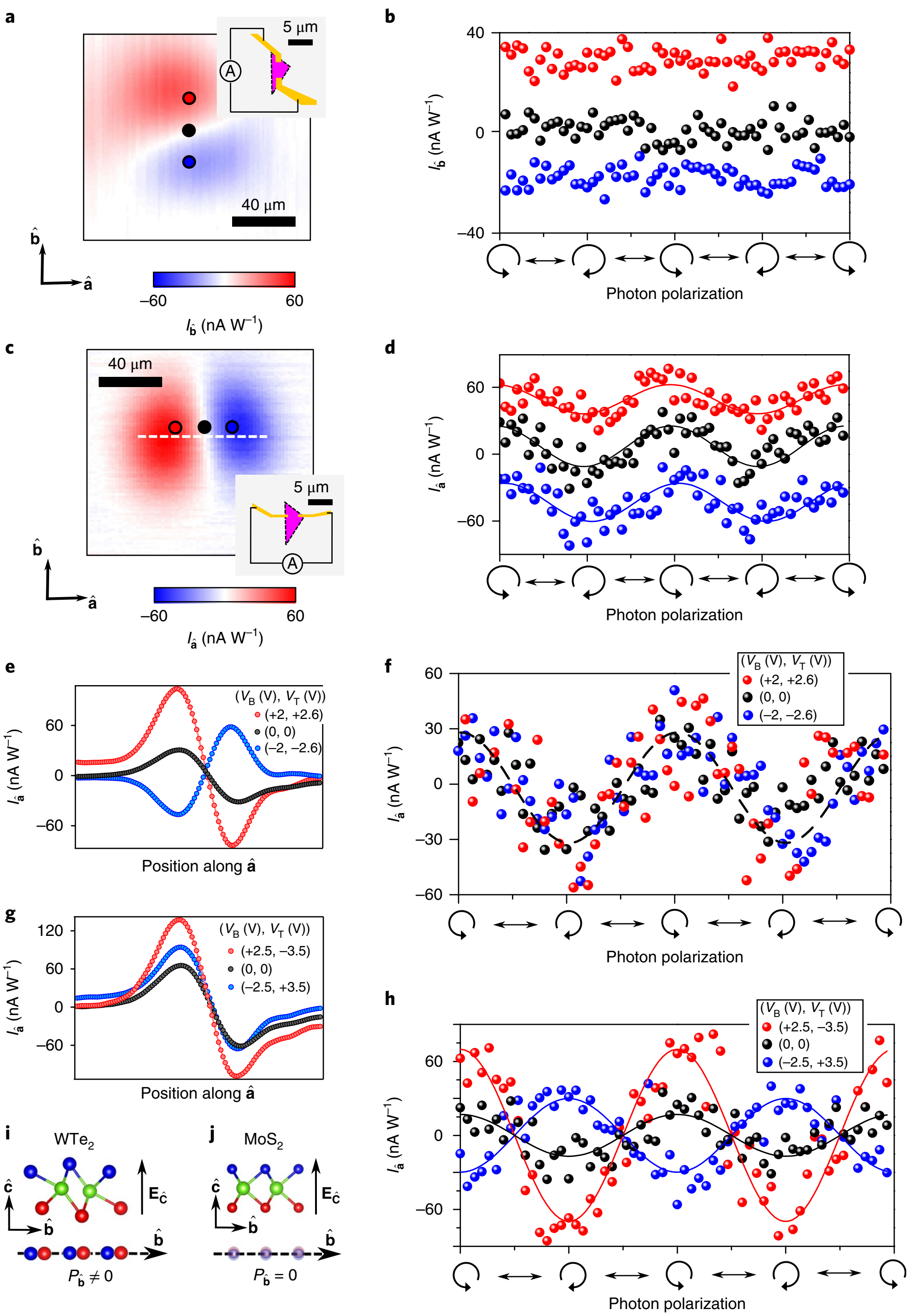

Fig. 2 | Observation of circular photogalvanic effect in monolayer WTe_2. **a**, Photocurrent along $\hat{\mathbf{b}}$ ($I_{\hat{b}}$) with linear polarized light as a function of the beam spot location in the $\hat{\mathbf{a}}$–$\hat{\mathbf{b}}$ plane. Inset: device schematics showing the two electrodes connected to the current amplifier. The spatial sizes of the photocurrent map and the device schematics are shown by the two scale bars, respectively. Each sample-to-contact interface serves as a thermal-electric junction. Photo-thermal currents can arise from both interfaces, leading to the bipolar spatial configuration. The black circle on the photocurrent map roughly corresponds to the centre of the WTe_2 sample, which has the same distance to the two contacts. **b**, Polarization-dependent $I_{\hat{b}}$ with the light spot fixed at the red, black and blue dots shown in **a**. **c**,**d**, Same as panels **a**,**b** but for the photocurrent along $\hat{\mathbf{a}}$ ($I_{\hat{a}}$). **e**,**f**, $I_{\hat{a}}$ as a function of laser spot position (**e**) along the dashed line in **c** or as a function of the polarization (**f**) for three different doping levels. **g**,**h**, Same as **e**,**f** but for three different displacement fields. All data in this figure were taken at $T = 150$ K. **i**,**j**, We colour the top and bottom atomic layers differently as the out-of-plane D causes the two layers to have different on-site potential energies. In MoS_2 (**j**), an out-of-plane D field only causes an out-of-plane (+$\hat{\mathbf{c}}$ to −$\hat{\mathbf{c}}$) polarity. In contrast, because the monolayer WTe_2 lattice features a tilted parallelogram (**i**), an out-of-plane D field can give rise to an in-plane polarity along $\hat{\mathbf{b}}$.

As seen in Fig. 2e,f, the CPGE remains unchanged within a relatively large charge density range ($|n| \leq 10^{13}\,\mathrm{cm}^{-2}$), whereas the photo-thermal current changes sign as one varies the doping from electron-like to hole-like. The observed small but nonzero CPGE in the absence of external displacement fields can be explained by two possibilities. First, the actual structure of monolayer WTe_2 is $1T_d$.

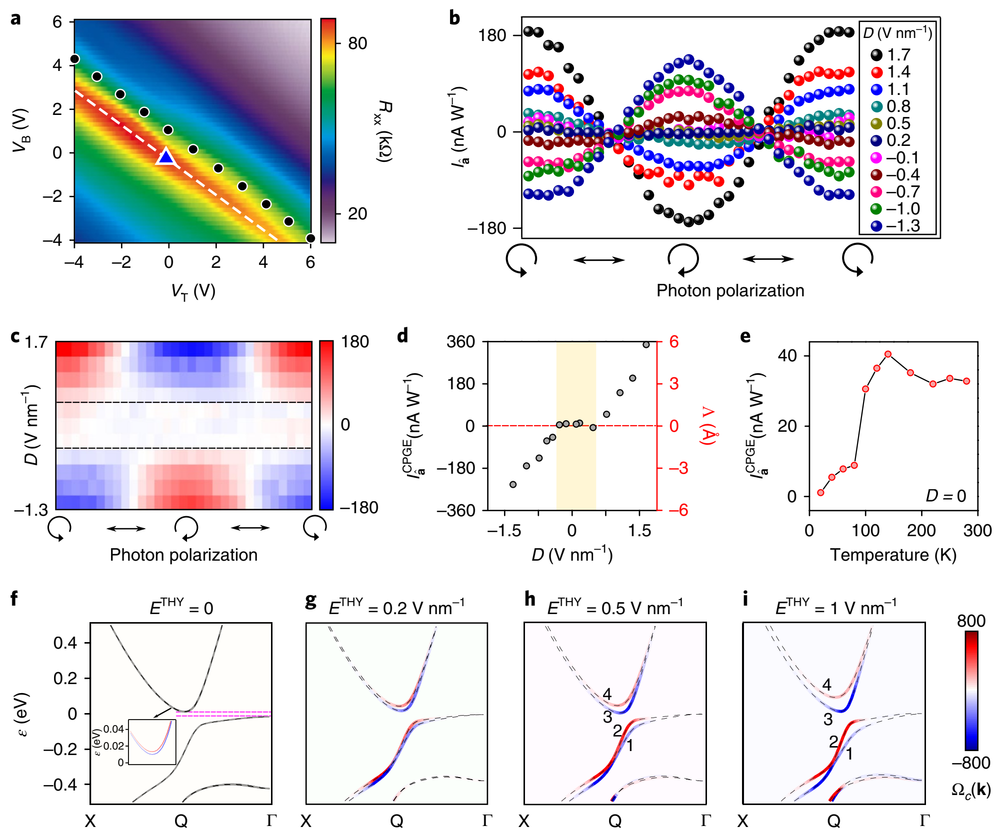

Fig. 3 | Systematic control of the circular photogalvanic current by displacement fields. a, Longitudinal DC resistance (R_{xx}) as a function of the top and bottom gate voltages at $T=20$ K. The dotted line defines the direction along which one can vary the displacement field while keeping the charge density invariant. **b**,**c**, Polarization-dependent circular photogalvanic effect (CPGE) currents for different displacement fields at $T=20$ K. **d**, Left-vertical axis shows the $I_{\hat{a}}^{CPGE}=I_{\hat{a}}(\mathrm{RCP})-I_{\hat{a}}(\mathrm{LCP})$ for right-circularly polarized (RCP) and left-circularly polarized (LCP) light as a function of the displacement field at $T=20$ K. Right-vertical axis shows our estimate of the corresponding Berry curvature dipole for each data point. **e**, Temperature-dependent $I_{\hat{a}}^{CPGE}$ in the absence of displacement fields. **f**–**i**, First-principles-calculated band structure and Berry curvature (represented by the blue-red colour) of monolayer WTe_2 using the Heyd–Scuseria–Ernzerhof (HSE) method (see the Methods section) for different out-of-plane electric fields. The four low-energy bands are labelled as 1 to 4. By fixing the hybrid parameter at HSE = 0.4, we obtain a global bandgap of ~30 meV (see also Supplementary Information VII.4 for previous studies), which leads to a minimum direct bandgap of ~150 meV without the displacement field. The minimum direct bandgap decreases back to less than 120 meV for $E \geq 0.2$ V nm^{-1} (**g**). The theoretical electric field (E^{THY}) here can be related to the displacement field by $D^{THY}=\varepsilon^{WTe_2}E^{THY}$

Second, the device has a small out-of-plane electric field coming from the dielectric environment[31]. To discern the two possibilities, one needs to perform structural measurements (for example, Raman spectroscopy, photocurrent spectroscopy, or second harmonic generation) on suspended WTe_2 monolayer samples.

We now study the dependence of the CPGE with an external, out-of-plane, displacement field **D**. As shown by the red curve in Fig. 2h, the application of a displacement field significantly increases the CPGE. In contrast, as we reverse the direction of the displacement field, the CPGE (blue curve) flips sign. Such electrical switching and sign-reversal have not been achieved homogeneously throughout a three-dimensional bulk system[25,27–29,32], as the electrical gating affects only the surface or interface region of a bulk crystal. The strong dependence of the in-plane CPGE with the out-of-plane **D** reveals a previously uncharacterized field effect in monolayer WTe_2. In graphene, MoS_2, and other conventional materials, an out-of-plane **D** causes only an out-of-plane ($+\hat{c}$ to $-\hat{c}$) polarity[17,18,31] (Fig. 2j). In contrast, because of the distinct crystal structure of monolayer WTe_2, which features a tilted parallelogram on the $\hat{b}-\hat{c}$ plane (Fig. 2i), an out-of-plane **D** can give rise to an in-plane polarity along $\hat{b}$ that modulates the in-plane CPGE. In Supplementary Information I.1 and I.2, we present additional photocurrent data and perform symmetry analysis of the observed CPGE with the displacement field.

We then study the CPGE at low temperatures ($T=20$ K). Guided by the gate map of the four-probe electrical resistance $R_{xx}(V_T, V_B)$ (Fig. 3a), we are able to tune the displacement field over a wide range while keeping the charge density invariant. As shown in Fig. 3b, the magnitude of the CPGE can be modulated over a very wide range over the displacement field; Its direction flips as one reverses the displacement field. These CPGE observations show that the displacement field can very effectively break the inversion and $\mathcal{C}_{2a}$ symmetries of the low-energy electronic states. In Fig. 3f–i, we show first-principles-calculated band structures of monolayer WTe_2. We see that both the band splittings and the Berry curvatures increase with increasing displacement field. Interestingly, the field effect on the electronic states has a significant momentum dependence: the band splitting and Berry curvatures are selectively strong near the inverted gap edge Q/Q′ but weak elsewhere in the Brillouin zone. In the next section of the manuscript, we will systematically study the relationship between the CPGE and the Berry curvatures. We note that the strong dependence of the CPGE on the displacement field, including sign-reversal, serves as crucial evidence of the intrinsic nature: any extrinsic effect should rely on extrinsic asymmetries and those do not change significantly with the displacement field (Supplementary Information VII.1). In addition, the temperature-dependent CPGE data also show interesting features: the CPGE at large **D** is strong at both low and high temperatures (Figs. 2h and

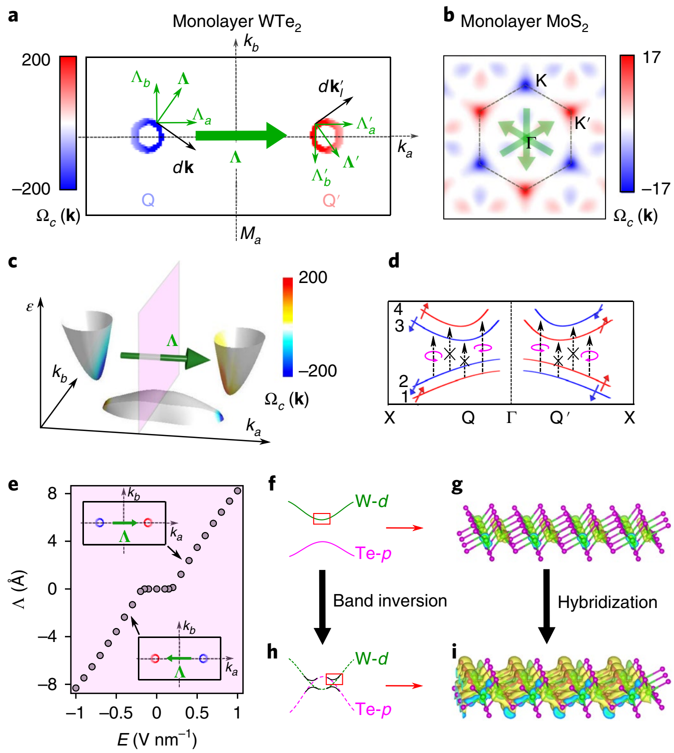

Fig. 4 | Berry curvature dipole and its control via a topological field effect. **a**,**c**, Berry curvature ($\Omega_c(\mathbf{k})$) of monolayer WTe_2 as a function of energy and momentum with the displacement field set at $E = +0.5\ \mathrm{V\,nm^{-1}}$ (same as Fig. 3i) in the $1T_d$ structure. **a**, $\Omega_c(\mathbf{k})$ of band 3 along the k-contours that correspond to an $\hbar\omega = 120$ meV inter-band transition. Note that k points that allow a fixed $\hbar\omega$ inter-band transition must be a closed contour. The small arrows depict the Berry curvature dipole contributed from two infinitesimal segments ($\mathrm{d}\mathbf{k}$, $\mathrm{d}\mathbf{k}'$) on the two contours. The big arrow shows the total Berry curvature dipole integrated over the contours (see equation 1). **b**, Berry curvature of monolayer MoS_2 as a comparison. **d**, Spin direction and optical selection rules between the lowest conduction and valence bands (1–4). **e**, Calculated Berry curvature dipole as a function of the electric displacement field. **f**,**g**, Calculated real-space distribution of the wavefunction amplitude at the inverted gap edge (Q point) in the presence of the band inversion. **h**,**i**, Same as in **f**,**g**, without the band inversion.

3b–d), whereas the CPGE at $\mathbf{D} = 0$ is weak at low temperatures and increases significantly at $T \sim 100$ K (Fig. 3e). These temperature-dependent CPGE data, along with the temperature-dependent DC transport data (Supplementary Information I.4), suggest that the bandgap decreases with increasing displacement field.

We now turn to the microscopic mechanism that generates the observed CPGE. We consider the scenario where the optical inter-band transition involves only the lowest conduction and highest valence bands. Under such conditions, we show below that the optical selection rules and the CPGE depend directly on the Berry curvature $\boldsymbol{\Omega}(\mathbf{k})$. In Figs. 4a,c and 3h, we find the following important characteristics for the Berry curvature of monolayer WTe_2 at a fixed field ($E^{\mathrm{THY}} = +0.5\ \mathrm{V\,nm^{-1}}$): $\Omega_c(\mathbf{k})$ exhibits clear hotspots near the Q and Q′ points where the direct energy gap is minimum and at all energies, the Berry curvature shows a bipolar configuration about the mirror plane $\mathcal{M}_a$. Inspired by the concept considered in refs. [11,33] for intra-band physics, we define an inter-band Berry curvature dipole ($\boldsymbol{\Lambda}^{\Omega}$) as

$$\boldsymbol{\Lambda}^{\Omega} = \oint \mathrm{d}\mathbf{k} \times \boldsymbol{\Omega}(\mathbf{k}) \qquad (1)$$

where the closed loop integral is defined along the contours in reciprocal space that correspond to a transition between bands 2 and 3 with energy $\hbar\omega$. Therefore, $\boldsymbol{\Lambda}^{\Omega}$ measures the degree of polarity of the Berry curvature texture on these contours (Fig. 4a). Combining equation 1 with the fact that $\Omega_c(k_a, k_b) = -\Omega_c(-k_a, k_b)$, enforced by the mirror plane $\mathcal{M}_a$, it is evident that $\Lambda_a^{\Omega} \neq 0, \Lambda_b^{\Omega} = 0$ for monolayer WTe_2. Remarkably, we found that the CPGE with normal incident light can be expressed concisely in terms of the Berry curvature dipole.

$$\mathbf{J}^{\mathrm{CPGE}} = \frac{e^3\tau}{\pi\hbar^2}\mathrm{Im}\left[\mathbf{E}(-\omega) \times \hat{\mathbf{c}}(\boldsymbol{\Lambda}^{\Omega} \cdot \mathbf{E}(\omega))\right] \qquad (2)$$

where τ is the relaxation time, $\mathbf{E}(\omega) = \frac{E_0}{\sqrt{2}}(\mathrm{e}^{i\omega t}, \mathrm{e}^{i(\omega t \pm \frac{\pi}{2})}, 0)$ describes normal incident right- and left-circularly polarized light. We see that, whereas the circular dichroic optical transition is possible as long as there is finite Berry curvature ($\mathcal{V}(\mathbf{k}) \propto \boldsymbol{\Omega}(\mathbf{k})$)[17,18,31], the CPGE (Supplementary Information VII.2) can occur in the presence of a Berry curvature dipole only when $\boldsymbol{\Lambda}^{\Omega} \neq 0$. The nonzero Λ_a^{Ω} in monolayer WTe_2 directly leads to the observed CPGE current along $\hat{\mathbf{a}}$ under normal incidence (see Supplementary Information VIII for discussions on the CPGE under oblique incidence). In Fig. 4e, we show the $\mathbf{D}$ dependence of the Berry curvature dipole Λ_a^{Ω}. For $|E| < 0.2\ \mathrm{V\,nm^{-1}}$, the direct bandgap is too large for an $\hbar\omega = 120$ meV transition; For $|E| \geq 0.2\ \mathrm{V\,nm^{-1}}$, $|\boldsymbol{\Lambda}^{\Omega}|$ increases monotonically with $|E|$. Importantly, both the trend and the order of magnitude of the calculated Berry curvature dipole (Fig. 4e) are consistent with the experimental data (Fig. 3d, see Supplementary Information VI for details).

We now study the origin of the observed field effect in monolayer WTe_2, which is in contrast to the very weak displacement field effect found in monolayer MoS_2[17,18,31] (see also Supplementary Information III). In the presence of a band inversion, the wavefunction spans all three atomic layers (Fig. 4h,i). As we remove the band inversion (Fig. 4f), the wavefunction becomes strongly localized near the central W atomic layer (Fig. 4g), and, at the same time, the field-effect-induced band splitting and Berry curvature become very small (see Supplementary Information II). Therefore, the field effect and the emergent Berry curvature hotspot near the inverted gap edge in monolayer WTe_2 are fingerprints of the topological band inversion.

To understand the role of spin in the circular polarization selection rule, we show that the spin polarizations of bands 2 and 3 (also 1 and 4) are along the same direction (Fig. 4e, see Supplementary Information IV for more details). The completely overlapping spin wavefunctions between bands 2 and 3 (Fig. 4e) demonstrate (from a different perspective) that the selection rule for circularly polarized light between these two bands arises purely from the Berry curvature. Moreover, within a constant energy contour, the spin texture shows a canted (between $\hat{\mathbf{b}}$ and $\hat{\mathbf{c}}$) Zeeman-like configuration on each constant energy contour (Supplementary Information IV), which is different from the out-of-plane Zeeman-like spin texture in MoS_2.

We highlight the key observations of monolayer $1T_d$ WTe_2 in comparison to graphene and MoS_2. The first observation is that the Berry curvature in monolayer WTe_2 is polar. In contrast, the Berry curvatures in unstrained MoS_2 and gapped graphene have a zero dipole and thus do not support the CPGE with normally incident light (Supplementary Information VII.5), which can also be seen from the following two factors: first, around each contour at K and K′, the Berry curvature is three-fold symmetric; second, there are three degenerate K–K′ pairs (Fig. 4b). We note that a nonzero Berry curvature dipole is allowed once the three-fold rotational symmetry is broken. Indeed, a recent experiment[34] showed evidence for it from current-induced magnetization in strained MoS_2. In addition, due to the inverted band structure, the Berry curvature in monolayer WTe_2 forms an intense hotspot, whose amplitude is more than one order of magnitude larger than that in MoS_2. The second observation is that the displacement field effect in monolayer WTe_2 is the first strong field effect in a monolayer crystal. This new field effect is a direct consequence of the topological band inversion and further shows an 'out-of-plane to in-plane' coupling. By contrast, previous field effects in MoS_2 and graphene are strong only in thicker layers[17,18,31], which mainly come from the coupling between different layers separated by the van der Waals gap.

Our results represent a clear experimental demonstration of a Berry curvature dipole, which can be controlled by electrical means. Such a tunable Berry curvature dipole not only leads to the electrically switchable CPGE observed here, but may enable a wide range of other quantum geometrical phenomena, such as magnetochiral[16], quantum nonlinear Hall[11], rectification[15], and orbital-Edelstein[12] effects. For all these phenomena, a nonzero Berry curvature is insufficient whereas a Berry curvature dipole is truly required. Many of these phenomena have not been observed in any solid state system and are therefore of great fundamental interest. Besides the above single-particle phenomena, the selection rules for circularly polarized light mean that monolayer WTe_2, analogous to MoS_2 and gapped graphene, can be used to realize chiral edge plasmons[13,14]. It is worth noting that monolayer WTe_2 has topological edge states, whose role in plasmon physics has not been studied, even theoretically. Further, the effect of a nontrivial Berry curvature and a Berry curvature dipole in gate-induced superconductivity[7] awaits exploration[21]. More broadly, the discovery of a Berry curvature dipole, along with the previous observations[2–8], establish monolayer WTe_2 as an extremely rich, atomically thin platform to explore topological physics, quantum geometrical physics, unconventional superconductivity and how these effects interplay with each other.

Methods

Methods, including statements of data availability and any associated accession codes and references, are available at https://doi.org/10.1038/s41567-018-0189-6.

Received: 6 November 2017; Accepted: 25 May 2018;
Published online: 2 July 2018

References

1. Qian, X., Liu, J., Fu, L. & Li, J. Quantum spin Hall effect in two-dimensional transition metal dichalcogenides. *Science* **346**, 1344–1347 (2014).
2. Zheng, F. et al. On the quantum spin Hall gap of monolayer 1T′–WTe_2. *Adv. Mater.* **28**, 4845–4851 (2016).
3. Fei, Z. et al. Edge conduction in monolayer WTe_2. *Nat. Phys.* **13**, 677–682 (2017).
4. Tang, S. et al. Quantum spin Hall state in monolayer 1T′-WTe_2. *Nat. Phys.* **13**, 683–687 (2017).
5. Jia, Z.-Y. et al. Direct visualization of a two-dimensional topological insulator in the single-layer 1T′-WTe_2. *Phys. Rev. B* **96**, 041108 (2017).
6. Wu, S. et al. Observation of the quantum spin Hall effect up to 100 kelvin in a monolayer crystal. *Science* **359**, 76–79 (2018).
7. Wu, S. et al. Observation of Topological Insulating and Superconducting Ground States of Monolayer WTe_2. *Bull. Am. Phys. Soc.* https://meetings.aps.org/Meeting/MAR18/Session/B37.4 (2018).
8. MacNeill, D. et al. Control of spin–orbit torques through crystal symmetry in WTe_2 ferromagnet bilayers. *Nat. Phys.* **13**, 300–305 (2016).
9. Nagaosa, N., Sinova, J., Onoda, S., MacDonald, A. & Ong, N. P. Anomalous Hall effect. *Rev. Mod. Phys.* **82**, 1539–1592 (2010).
10. Moore, J. E. & Orenstein, J. Confinement-induced Berry phase and helicity-dependent photocurrents. *Phys. Rev. Lett.* **105**, 026805 (2010).
11. Sodemann, I. & Fu, L. Quantum nonlinear Hall effect induced by Berry curvature dipole in time-reversal invariant materials. *Phys. Rev. Lett.* **115**, 216806 (2015).
12. Yoda, T., Yokoyama, T. & Murakami, S. Orbital Edelstein effect as a condensed-matter analog of solenoid. *Nano Lett.* **18**, 916–920 (2018).
13. Basov, D., Fogler, M. & de Abajo, F. G. Polaritons in van der Waals materials. *Science* **354**, aag1992 (2016).
14. Low, T. et al. Polaritons in layered two-dimensional materials. *Nat. Mater.* **16**, 182–194 (2017).
15. Ideue, T. et al. Bulk rectification effect in a polar semiconductor. *Nat. Phys.* **13**, 578–583 (2017).
16. Rikken, G. & Raupach, E. Observation of magneto-chiral dichroism. *Nature* **390**, 493–494 (1997).
17. Xu, X., Yao, W., Xiao, D. & Heinz, T. F. Spin and pseudospin in layered transition metal dichalcogenides. *Nat. Phys.* **10**, 343–350 (2014).
18. Mak, K. F. & Shan, J. Photonics and optoelectronics of 2D semiconductor transition metal dichalcogenides. *Nat. Photon.* **10**, 216–226 (2016).
19. de Juan, F., Grushin, A. G., Morimoto, T. & Moore, J. E. Quantized circular photogalvanic effect in Weyl semimetals. *Nat. Commun.* **8**, 15995 (2017).
20. Son, D. & Spivak, B. Chiral anomaly and classical negative magnetoresistance of Weyl metals. *Phys. Rev. B* **88**, 104412 (2013).
21. Gradhand, M. & Annett, J. F. The Berry curvature of the Bogoliubov quasiparticle Bloch states in the unconventional superconductor Sr_2RuO_4. *J. Phys. Condens. Matter* **26**, 274205 (2014).
22. Ali, M. N. et al. Large, non-saturating magnetoresistance in WTe_2. *Nature* **514**, 205–211 (2014).
23. Soluyanov, A. et al. Type-II Weyl semimetals. *Nature* **527**, 495–498 (2015).
24. Sipe, J. & Shkrebtii, A. Second-order optical response in semiconductors. *Phys. Rev. B* **61**, 5337–5352 (2000).
25. Wittmann, B. et al. Circular photogalvanic effect in HgTe/CdHgTe quantum well structures. *Semicond. Sci. Technol.* **25**, 095005 (2010).
26. Hosur, P. Circular photogalvanic effect on topological insulator surfaces: Berry-curvature-dependent response. *Phys. Rev. B* **83**, 035309 (2011).
27. McIver, J., Hsieh, D., Steinberg, H., Jarillo-Herrero, P. & Gedik, N. Control over topological insulator photocurrents with light polarization. *Nat. Nanotech.* **7**, 96–100 (2012).
28. Yuan, H. et al. Generation and electric control of spin–valley-coupled circular photogalvanic current in WSe_2. *Nat. Nanotech.* **9**, 851–857 (2014).
29. Ma, Q. et al. Direct optical detection of Weyl fermion chirality in a topological semimetal. *Nat. Phys.* **13**, 842–847 (2017).
30. Muechler, L., Alexandradinata, A., Neupert, T. & Car, R. Topological nonsymmorphic metals from band inversion. *Phys. Rev. X* **6**, 041069 (2016).
31. Wu, S. et al. Electrical tuning of valley magnetic moment through symmetry control in bilayer MoS_2. *Nat. Phys.* **9**, 149–153 (2013).
32. Dhara, S., Mele, E. J. & Agarwal, R. Voltage-tunable circular photogalvanic effect in silicon nanowires. *Science* **349**, 726–729 (2015).
33. Zhang, Y., Sun, Y. & Yan, B. The Berry curvature dipole in Weyl semimetal materials: an *ab initio* study. *Phys. Rev. B* **97**, 041101(R) (2017).
34. Lee, J., Wang, Z., Xie, H., Mak, K. & Shan, J. Valley magnetoelectricity in single-layer MoS_2. *Nat. Mater.* **16**, 887–891 (2017).

Acknowledgements

We acknowledge Y. Lin and T. Palacios for their assistance on measurements. Work in the P.J.-H. group was partly supported by the Center for Excitonics, an Energy Frontier Research Center funded by the US Department of Energy (DOE), Office of Science, Office of Basic Energy Sciences (BES) under Award Number DESC0001088 (fabrication and measurement) and partly through AFOSR grant FA9550-16-1-0382 (data analysis), as well as the Gordon and Betty Moore Foundation's EPiQS Initiative through Grant GBMF4541 to P.J.-H. This work made use of the Materials Research Science and Engineering Center Shared Experimental Facilities supported by the National Science Foundation (NSF) (Grant No. DMR-0819762). N.G. and S.Y.X. acknowledge support from DOE, BES DMSE (data taking and analysis), and the Gordon and Betty Moore Foundations EPiQS Initiative through Grant GBMF4540 (manuscript writing). The WTe_2 crystal growth performed at Princeton University was supported by an NSF MRSEC grant, DMR-1420541 (Q.D.G. and R.J.C.). K.W. and T.T. acknowledge support from the Elemental Strategy Initiative conducted by the MEXT, Japan and JSPS KAKENHI Grant Numbers JP15K21722. H.S. and L.F. acknowledge support from NSF Science and Technology Center for Integrated Quantum Materials grant DMR-1231319 (theory). The WTe_2 film growth performed at Nanyang Technological University was supported by MOE Tier 2 grants MOE2016-T2-2-153 and MOE2015-T2-2-007 and the Singapore National Research Foundation under NRF award number NRF-NRFF2013-08. T.R.C. was supported by the Ministry of Science and Technology under the MOST Grant for the Columbus Program NO. 107-2636-M-006-004, National Cheng Kung University, Taiwan, and National Center for Theoretical Sciences (NCTS), Taiwan.

Author contributions

Q.M. and S.Y.X. performed the measurements and analysed the data with help from A.M.M.V. V.F. and S.W. fabricated the devices. Q.D.G. and R.J.C. grew the bulk WTe_2 single crystals. K.W. and T.T. grew the bulk hBN single crystals. J.Z. and Z.L. grew the WTe_2 CVD films. T.R.C., G.C. and H.L. calculated the first-principles band

structures. S.Y.X., Q.M., H.S. and L.F. did theoretical analysis of the Berry curvature dipole with help from C.K.C. H.S. and L.F. performed analysis of the $\mathbf{k} \cdot \mathbf{p}$ model. S.Y.X. and Q.M. wrote the manuscript with input from all authors. P.J.-H. and N.G. supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41567-018-0189-6.

Reprints and permissions information is available at www.nature.com/reprints.

Correspondence and requests for materials should be addressed to N.G. or P.J.

Publisher's note: Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Methods

Device fabrication for exfoliated flake samples. Our fabrication of the dual-gated monolayer WTe_2 devices consists of two phases. Phase I was done under ambient conditions: local bottom PdAu gates were first defined on the standard Si/SiO_2 substrates. A suitable hexagonal hexagonal BN (hBN) flake was exfoliated onto a separate Si/SiO_2 substrate, then picked up using a polymer-based dry transfer technique and placed onto the pre-patterned local bottom gate. Electrical contacts (PdAu, ~20 nm thick) were deposited onto the bottom hBN flake with electron-beam lithography and metal deposition. Phase II was done fully inside the glovebox with an argon environment. Monolayer WTe_2 flakes were exfoliated from a bulk crystal onto a Si/SiO_2 chip. Thin graphite (as top gate electrode, ~10 nm), hBN (~10 nm) and monolayer WTe_2 were sequentially picked up and then transferred onto the local bottom gate/hBN/contact substrate. Extended leads connecting the top gate graphene to wire bonding pads were pre-made together with the metal contacts in Phase I. In such a dual-gated device, the charge density can be obtained by $n = \frac{\epsilon_0 \epsilon^{\mathrm{hBN}}}{e}(V_\mathrm{T}/h_\mathrm{T} + V_\mathrm{B}/h_\mathrm{B})$. The displacement field is determined by $\mathbf{D} = (\epsilon^{\mathrm{hBN}} V_\mathrm{T}/h_\mathrm{T} - \epsilon^{\mathrm{hBN}} V_\mathrm{B}/h_\mathrm{B})/2$[35]. Here n is the charge density, $\mathbf{D}$ is the externally applied displacement field, V_T (V_B) are the gate voltages, and $\epsilon^{\mathrm{hBN}} = 3$, h_T=10 nm, and h_B=8 nm are the relative dielectric constant and the thicknesses of the top (T) and bottom (B) hBN layers of the presented device, respectively.

Growth method and device fabrication for CVD films. The mixed compounds (WO_3: WCl_6: Te = 1:1:1) were placed in the boat at the middle position of the tube. The Te powder will reduce the melting point of the compounds. 0.5 g Te powder is placed on the upstream, which is used to maintain the Te atmosphere during the whole reaction process. WTe_2 monolayer layers can be found on a SiO_2 (285 nm)/Si wafer at the downstream of the quartz tube. The growth temperature is between 750–850 °C. The samples were immediately covered by poly(methyl methacrylate) (PMMA) for protection against oxidization. Electrical contacts (PdAu, ~20 nm thick) were deposited onto chemical vapour deposited (CVD) film with electron-beam lithography and metal deposition. The SiO_2 layer also served as the global back gate dielectric.

Mid-infrared scanning photocurrent microscopy. The fabricated device was wire-bonded onto a chip carrier and placed in an optical scanning microscope set-up that combines electronic transport measurements with light illumination. The laser source is a temperature-stablized CO_2 laser ($\lambda = 10.6\,\mu$m, $\hbar\omega = 120$ meV). A focused beam spot (diameter $d \simeq 50\,\mu$m) is scanned (using a two-axis piezo-controlled scanning mirror) over the entire sample and the current is recorded at the same time to form a colour map of the photocurrent as a function of spatial positions. Reflected light from the sample is collected to form a simultaneous reflection image of the sample. The absolute location of the photo-induced signal is therefore found by comparing the photocurrent map to the reflection image. The light is first polarized by a polarizer and the chirality of light is further modulated by a rotatable quarter-wave plate.

First-principles calculations. First-principles calculations were performed by the OPENMX code within the framework of the generalized gradient approximation (GGA) of density functional theory[36,37]. As the GGA sometimes underestimates the bandgap, the Heyd–Scuseria–Ernzerhof (HSE) method was employed to adjust the bandgap according to the previous photoemission results.

The connection between Berry curvature and the CPGE. Here we briefly describe the connection between the Berry curvature dipole and the CPGE current. Supplementary Information V presents a detailed derivation. The generation of the CPGE with normal incident light consists of two steps: first, circularly polarized light excites inter-band transitions with an optical selection rule; second, the optically excited electrons and holes travel at their respective group velocities, leading to a nonzero CPGE current. In the case where only two bands participate in the inter-band process, it has been theoretically shown[19,31,38] that the difference between the inter-band transition rate for right- and left-circularly polarized light, $\mathcal{V}(\mathbf{k}) = |\mathcal{P}^{\mathrm{RCP}}(\mathbf{k})|^2 - |\mathcal{P}^{\mathrm{LCP}}(\mathbf{k})|^2$, is directly proportional to the Berry curvature, that is, $\mathcal{V}(\mathbf{k}) \propto \Omega(\mathbf{k})$. Therefore, the inter-band transition near Q and Q′ in monolayer WTe_2 selects opposite circularly polarized light because of their opposite Berry curvatures. Under this condition, the CPGE current can be expressed as $\mathbf{J}^{\mathrm{CPGE}} = \frac{e^3\tau}{\pi\hbar^2}\mathrm{Im}[\mathbf{E}(-\omega) \times \hat{\mathbf{c}}(\mathbf{\Lambda}^{\Omega} \cdot \mathbf{E}(\omega))]$.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

References

35. Taychatanapat, T. & Jarillo-Herrero, P. Electronic transport in dual-gated bilayer graphene at large displacement fields. *Phys. Rev. Lett.* **105**, 166601 (2010).
36. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
37. Heyd, J., Scuseria, G. E. & Ernzerhof, M. Hybrid functionals based on a screened Coulomb potential. *J. Chem. Phys.* **118**, 8207–8215 (2003).
38. Yao, W., Xiao, D. & Niu, Q. Valley-dependent optoelectronics from inversion symmetry breaking. *Phys. Rev. B* **77**, 235406 (2008).